

In[*]:= Quit

In[*]:= \$Assumptions = Dif > 0 && t > 0

Out[*]=

Dif > 0 && t > 0

In[*]:= EFP = D[f[t, x], t] + μ D[f[t, x], x] - Dif D[f[t, x], x, x]

Out[*]=

$\mu f^{(0,1)}[t, x] - \text{Dif } f^{(0,2)}[t, x] + f^{(1,0)}[t, x]$

In[*]:= $f[t_ , x_] = \frac{1}{(4 \pi \text{Dif } t)^{1/2}} \text{Exp}\left[-\frac{(x - x0 - \mu t)^2}{4 \text{Dif } t}\right]$

Out[*]=

$$\frac{e^{-\frac{(x-x0-\mu t)^2}{4 \text{Dif } t}}}{2 \sqrt{\pi} \sqrt{\text{Dif } t}}$$

Checkeo de normalización: La probabilidad de que la partícula esté en algún lugar debe ser 1

In[*]:= Integrate[f[t, x], {x, -Infinity, +Infinity}]

Out[*]=

1

Checkeo de que es solución

In[*]:= EFP // FullSimplify

Out[*]=

0

Promedio

In[*]:= Integrate[x f[t, x], {x, -Infinity, +Infinity}]

Out[*]=

$x0 + t \mu$

Desviación estandar² Calculamos $\langle x^2 \rangle - \langle x \rangle^2$

In[*]:= Integrate[x² f[t, x], {x, -Infinity, +Infinity}] - (Integrate[x f[t, x], {x, -Infinity, +Infinity}])²

Out[*]=

2 Dif t

In[*]:= f[t, x] /. {x0 → 0, Dif → 1, μ → 0}

Out[*]=

$$\frac{e^{-\frac{x^2}{4t}}}{2 \sqrt{\pi} \sqrt{t}}$$

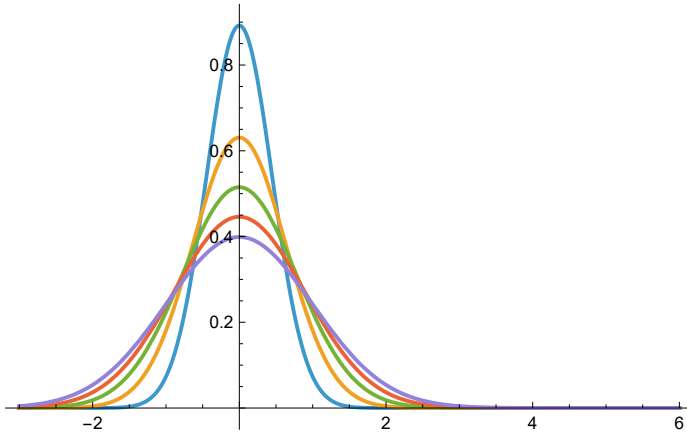
In[*]:= Table $\left[\frac{e^{-\frac{x^2}{4t}}}{2 \sqrt{\pi} \sqrt{t}}, \{t, 0.1, 0.5, 0.1\}\right]$

Out[*]=

$\{0.892062 e^{-2.5 x^2}, 0.630783 e^{-1.25 x^2}, 0.515032 e^{-0.833333 x^2}, 0.446031 e^{-0.625 x^2}, 0.398942 e^{-0.5 x^2}\}$

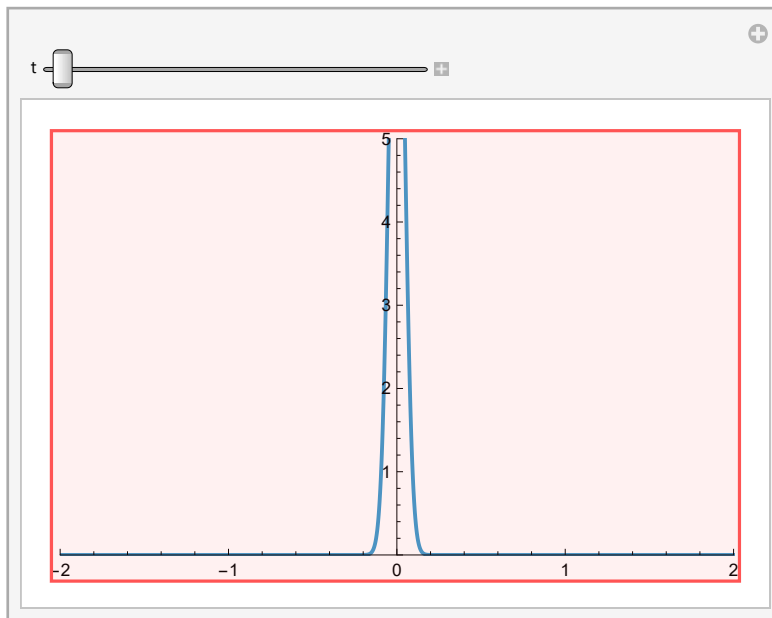
```
In[ ]:= Plot[{0.8920620580763855` e-2.5` x2,
0.6307831305050401` e-1.25` x2, 0.5150322693642527` e-0.8333333333333333` x2,
0.44603102903819275` e-0.625` x2, 0.39894228040143265` e-0.5` x2}, {x, -3, 6}]
```

Out[]:=



```
In[ ]:= Manipulate[Plot[ $\frac{e^{-\frac{x^2}{4t}}}{2\sqrt{\pi}\sqrt{t}}$ , {x, -2, 2}, PlotRange -> {{-2, 2}, {0, 5}}], {t, 1/1000, 1/10}]
```

Out[]:=



General: Exp[-999.918] is too small to represent as a normalized machine number; precision may be lost. [i](#)

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In[]:= Quit

Fotos de la distribución a distintos tiempos, para $v=1$, $d=1$, $x_0=1$

In[]:= {v = 1, d = 1, x0 = 1}

Out[]:=

{1, 1, 1}

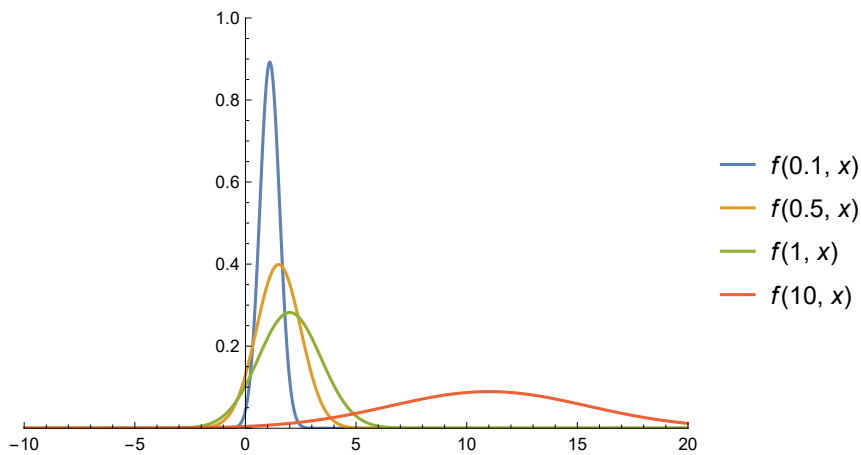
In[]:= $f[t_, x_] = \frac{1}{(4 \pi d t)^{1/2}} \text{Exp}\left[\frac{-(x - x_0 - v t)^2}{4 d t}\right]$

Out[]:=

$$\frac{e^{-\frac{(-1-t+x)^2}{4t}}}{2 \sqrt{\pi} \sqrt{t}}$$

In[]:= Plot[{f[0.1, x], f[0.5, x], f[1, x], f[10, x]}, {x, -10, 20},
PlotRange -> {{-10, 20}, {0, 1}}, PlotLegends -> "Expressions"]

Out[]:=



In[]:= Quit

```

(*Parameters*) d = 1;
x0 = 0;
v = 0;

f[t_, x_] := (1 / (4 π d t) ^ (1 / 2)) Exp[-(x - x0 - v t) ^ 2 / (4 d t)]

times = {0.5, 1.0, 10, 100};
colors = {RGBColor[0.2, 0.4, 0.8],
  RGBColor[0.8, 0.2, 0.2], RGBColor[0.1, 0.6, 0.3], RGBColor[0.7, 0.3, 0.8]};

(*Integrate the right tail from x=2 to infinity*)
areas = Table[NIntegrate[f[t, x], {x, 2, Infinity}], {t, times}];

plots = Table[Module[{ymax, shadedPts, shadedPoly},
  ymax = 1.1 Max[Table[f[times[[i]], x], {x, -5, 10, 0.05}]];
  (*Sample the tail from x=2 to x=10 (effectively infinity for display)*)
  shadedPts = Table[{x, f[times[[i]], x]}, {x, 2, 10, 0.05}];
  shadedPoly = Polygon[Join[{{2, 0}}, shadedPts, {{10, 0}}]];
  Show[
    Plot[{}], {x, -5, 10}, PlotStyle -> {colors[[i]], Thickness[0.003]}, PlotRange -> {{-5, 10},
      {0, ymax}}, (*Tail polygon shading*) Graphics[{Opacity[0.5],
      colors[[i]], shadedPoly, Opacity[1], Dashed, colors[[i]], Line[
      {{2, 0}, {2, f[times[[i]], 2}]}]}], Frame -> True, FrameLabel -> {"f(t, x)", None},
      {"x", Style["t = "<>ToString[times[[i]]], 14, Bold, colors[[i]]]}],
    GridLines -> Automatic, GridLinesStyle -> Directive[GrayLevel[0.85], Thin],
    PlotLabel -> Style["P(x > 2) = "<>ToString[NumberForm[areas[[i]], {4, 4}]], 12,
      Bold, colors[[i]]], Background -> GrayLevel[0.97], ImageSize -> 280], {i, 1, 4}];

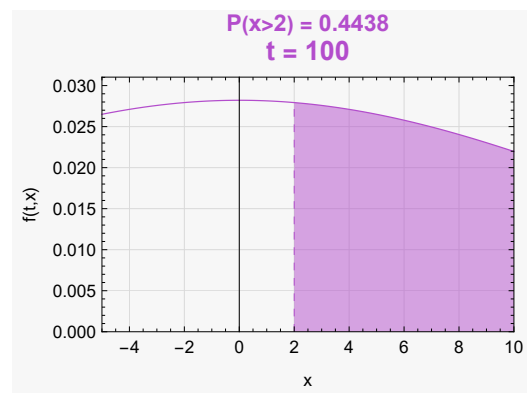
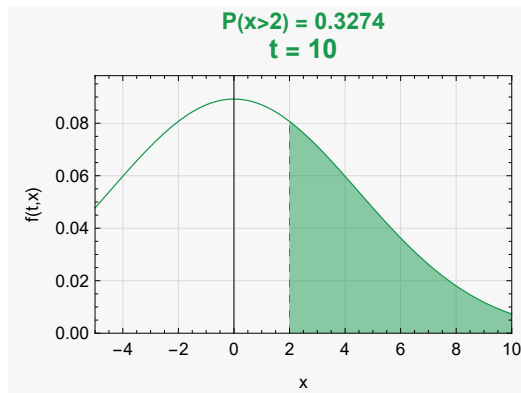
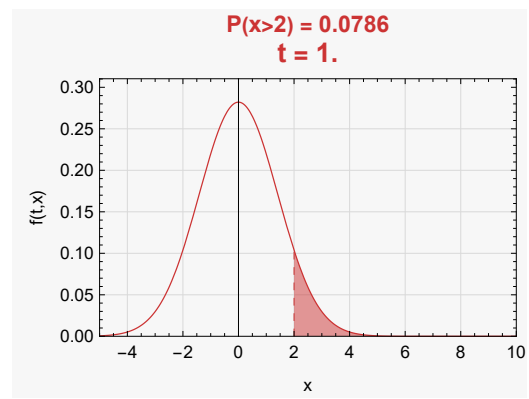
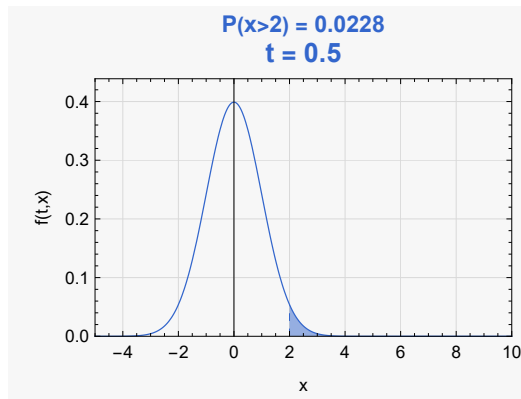
table = Grid[Join[{{Style["t", Bold, 13], Style["Peak Position", Bold, 13],
  Style["Peak Height", Bold, 13], Style["P(x > 2)", Bold, 13]}},
  Table[{Style[times[[i]], colors[[i]], 12], Style[NumberForm[x0 + v times[[i]],
    {4, 3}], 12], Style[NumberForm[1 / (4 π d times[[i]]) ^ (1 / 2), {4, 4}], 12],
    Style[NumberForm[areas[[i]], {4, 5}], colors[[i]], Bold, 12]}, {i, 1, 4}]],
  Frame -> All, FrameStyle -> GrayLevel[0.7], Spacings -> {2, 1},
  Background -> {{}, {GrayLevel[0.88], GrayLevel[0.97],
    GrayLevel[0.97], GrayLevel[0.97], GrayLevel[0.97]}}];

Column[{Style["Diffusion Distribution f(t, x)", 18, Bold, GrayLevel[0.2]],
  Style["Parameters: d = "<>ToString[d]<>, x0 = "<>ToString[x0]<>
  ", v = "<>ToString[v], 12, GrayLevel[0.5]], Spacer[10],
  GraphicsGrid[{plots[[{1, 2}]], plots[[{3, 4}]]}, Spacings -> {10, 10}],
  Spacer[10], table}, Alignment -> Center]

```

Out[8]=

Diffusion Distribution $f(t,x)$

Parameters: $d = 1$, $x_0 = 0$, $v = 0$ 

t	Peak Position	Peak Height	$P(x > 2)$
0.5	0.000	0.3989	0.02275
1.	0.000	0.2821	0.07865
10	0	$\frac{1}{2\sqrt{10}\pi}$	0.32740
100	0	$\frac{1}{20\sqrt{\pi}}$	0.44380